

YZV202E Optimization for Data Science

Project Final Report

Traffic Congestion Prediction Optimization

Anıl Dervişoğlu
Artificial Intelligence and Data Engineering
Istanbul Technical University
dervisoglua22@itu.edu.tr
150220344

Ömer Faruk Zeybek
Artificial Intelligence and Data Engineering
Istanbul Technical University
zeybeko22@itu.edu.tr
150220743

Abstract—The aim of this project is to predict daily traffic congestion in Istanbul by training an LSTM model on a dataset that combines traffic data, bus and train stations, population density, weather conditions, and district data. We tested four optimization algorithms (Adam, Bayesian Optimization, Particle Swarm Optimization (PSO), and gradient-based Hybrid PSO) and compared them in terms of performance and computational efficiency. Hybrid PSO yields the lowest mean squared error but is slower. Adam achieves nearly the same performance but provides much faster results, vanilla PSO falls in the middle, and Bayesian Optimization shows the worst performance in our data. These results demonstrate that selecting an optimization algorithm requires striking a clear balance between performance and efficiency.

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I. PROJECT DESCRIPTION

With the increasing population and crowded traffic in Istanbul, the aim of this project is to understand the historical data and predict a self made traffic congestion level showing

a value between $[0, 1]$ range indicating how dense the traffic actually is.

Following our motivation, we also decided to understand and explore the harder path, optimization in neural networks. To do so we have analyzed several models as to understand which would fit the limited data we have in hand the best. And with input parameters of the specific day of the week, the district, whether it is a special day, population in the district, the weather, number of railway and bus stations in 1 km radius; as the data we had were in time series format and we wanted also use the information from the past inputs, we decided to go on with LSTM model.

For the sustainability area, our project is directly related with public health with the increased air pollution, stress (people might get angry), less physical activity and so on depending on the traffic congestion. It also affects the country economically as it reduces immediate access to many services. With our project, by predicting congestion accurately, we expect to support the "sustainable" decision making in transportation.

II. PROBLEM DEFINITION

Formally, the problem we tried to solve in this project is to understand and predict the traffic congestion level given specific day, location and many other parameters described below with a value between $[0, 1]$.

Below, each input parameter and their specific range can be seen also with their abbreviations to use later in the proposal, $t \geq 0$ stands for the time step while x_n stands for the input parameters:

$$\begin{aligned} \text{is_special_day} &= s \in \{0, 1\} = x_1^t \\ \text{grid_population} &= p \geq 0 = x_2^t \\ \text{\#_of_bus_stops} &= bs \geq 0 = x_3^t \\ \text{\#_of_railway_stops} &= rs \geq 0 = x_4^t \\ \text{day_of_week} &= dw \in \{1, 2, \dots, 7\} = x_5^t \\ \text{district} &= d \in \{0, 1, \dots, D\} = x_6^t \\ \text{weather_condition} &= w \in \{0, 1, 2, 3\} = x_7^t \end{aligned}$$

We can formally represent the input vector and output as follows:

$$X_t = [x_1^t, x_2^t, \dots, x_7^t]$$

$$y_t \in [0, 1]$$

The objective function (LSTM) can be then represented as (where k is the number of time steps used):

$$\hat{y}_t = f(X_{t-k}, X_{t-k+1} \dots X_t)$$

and the loss function (Mean Squared Error and Mean Absolute Error) for using our optimization functions is as follows:

$$L(\hat{y}, y)$$

We chose to first apply Adam and Bayesian as the optimization method to see the affects. Then, we also tried a combination of Particle Swarm Optimization with gradient as another addition alongside the direction, personal best and global best. For better comparison and understanding we compared our results with each of them and try to understand the trade offs.

III. DATASET

We chose six different datasets to complete our project. Descriptions of the datasets are given below:

- **Hourly Traffic Density Dataset**
 - This dataset contains hourly Istanbul location density and traffic information.
 - **Purpose of the dataset:** This dataset contains both location and hourly data. It also shows the average, maximum and minimum speed and the number of vehicles. This allows us to better predict traffic congestion.
 - **Data Source:** Ulusal Akıllı Şehir Platformu
- **IETT Bus Stops Data**
 - This dataset contains vector data of IETT bus stops.
 - **Purpose of the dataset:** Since buses are also traffic generators, it is reasonable to expect higher traffic density around bus stops, especially within a 1 km radius.
 - **Data Source:** Ulusal Akıllı Şehir Platformu
- **Rail System Station Points Data**
 - This dataset contains vector data of rail system station points.
 - **Purpose of the dataset:** There is a similar logic that traffic congestion will be greater near bus stops, but since subways are not subject to traffic, we believe that they can reduce traffic congestion in the area where they are located.
 - **Data Source:** Ulusal Akıllı Şehir Platformu
- **Population Density Dataset**
 - This dataset shows the density of population by region.
 - **Purpose of the dataset:** Using this dataset, which includes the population density in the districts of

Istanbul, we can predict that traffic will be higher on roads with higher population density and lower on roads with lower population density.

– **Data Source:** TÜİK

- **Weather Condition Dataset (External Data)**

- This dataset contains weather information for Istanbul from 2020 to 2025.

- **Purpose of the dataset:** Weather conditions are one of the most important factors in traffic density, this dataset contains weather information on a daily basis. We know that traffic congestion is higher when it is rainy, snowy and foggy. With this dataset, we can get better results.

- **Data Source:** Visual Crossing

- **Istanbul Districts Geojson (External Data)**

- Contains districts map of İstanbul in GeoJSON format.

- **Purpose of the dataset:** We used this data set to find out which district the roads and locations with latitude and longitude information belong to.

- **Data Source:** GitHub

IV. METHODOLOGY

A. Data Preparation:

We selected some set of features that are important and necessary for our model. Features include month of the year, day of the week, weather conditions, district population data and indicators for public holidays or special events, as well as traffic congestion levels from previous periods, and especially metro and bus stations near roads within a 1 km radius. After feature selection, we normalized the continuous variables in our data to achieve efficient training and convergence.

It would also be meaningful to detail some of the feature engineering done on the datasets. First and foremost, as we couldn't use data from all over the Istanbul as latitude and longitude information, we reduced them into one of the 38 districts they belong to using the Istanbul Districts Geojson data. But this was not enough. The data we had was designed on hourly data and we transformed it to daily data to reduce the size of it and training time as we only work on commodity hardware (more explicitly, our laptops). Aside from these, we get all the data into a daily format whether it was given in hours or as static information (like population or number of stations).

One other important data was our output variable, congestion value. To calculate it, we have utilized our data of *average speed* and *number of vehicles* in specific location in a given time. We combined them with the following logic:

- Removing the outliers outside of %5 and %95 quantiles for both speed and number of vehicles.
- Normalizing these values between $[0, 1]$.
- Taking the inverse of normalized average speed as we are trying to calculate how congested the traffic would be, ($inv_s = 1 - s$).

- Taking their average to calculate congestion value:

$$\text{congestion} = \#_vehicles/2 + inv_speed/2$$

B. Model Architecture:

We used LSTM in our project as we are dealing with time series prediction. Given an input of k time steps in the form of X_t feature vectors, we provided the input features we have to the model. Our base model contains an LSTM layer that learns the relationships within data. Also, to have an output $\hat{y}_t$ within the given $[0, 1]$ range we used a sigmoid activation function as to output. While we didn't specifically put out experimental results, we have tried different number of hidden dimensions and 32 worked out just fine for us. Also as the main topic of this lesson is optimization, we rather tried to keep a stationary value for these hyperparameters and focus on the optimization techniques as in the following sections.

C. Optimization Techniques:

At first, we tried Bayesian Optimization as we thought it was cost effective compared to many other methods and try to see the effects of a gradient free method on our model. To compare and later improve on we also tried Particle Swarm Optimization as another gradient free optimization method.

On another hand, we also tried using a hybrid approach using PSO with an additional parameter of gradient for each individual particle, we expected this approach to have quite the cost but our main idea was to see if we could find global optimums rather than local approximations and have a better model in the end. Later, as detailed in the experimental section, we have implemented many variations of it to both improve the performance and to decrease the time taken.

Finally, knowing that Adam is currently more favorite than any of our gradient free method, we also implemented such a model with it and compared the results.

D. Evaluation Metrics:

As explained in the problem definition, we evaluated our metrics generally with Mean Squared Error, but due to the nature of our data being in a range between $[0, 1]$ we also used in mean absolute error in our approaches to test the differences. Having quite the similar results, we have decided to give error results in this report using MSE.

V. REAL WORLD APPLICATION

Predicting traffic congestion accurately is really important for effective transportation management especially in crowded cities. LSTM model and the optimization techniques that we presented in this study can be applied in different real-world scenarios:

- **Smart City Infrastructure:** Authorities can use real-time traffic forecasts to make better decisions about traffic lights and control traffic congestion more effectively, helping traffic flow more efficiently.
- **Navigation Systems:** Real-time traffic congestion predictions can be integrated into navigation apps and GPS systems to recommend the most optimal routes.

- **Environmental Impact:** With this project we can optimize traffic flow minimize traffic congestion and contribute to reducing vehicle emissions and improving urban air quality.

Basically we can say that the methods and results of this project can help build smarter and more sustainable cities by supporting data-driven traffic management and planning.

VI. EXPERIMENTAL EVALUATION

A. Particle Adjustments:

While we had tried out different methods on gradient based and gradient free optimization methods on LSTM architecture, we also experimented alternative techniques to improve the quality of experiment. With this purpose, we have implemented the following methods on our Hybrid PSO optimization method:

- **Gradient Momentum:** As explained in the methodology, we have used gradient information on top of already existing information within each particle of swarm (random initial weight, personal best and global best). As we have learned from our lessons, it can be meaningful to use the previously calculated gradient value to calculate the current one.
- **Particle Momentum:** Rather than applying the idea of momentum only on gradient values, we wanted to generalize it and apply it on the whole particle velocity. This way the previous moves of the particle would give a slight effect in the current move.
- **Decreasing Learning Rate:** The idea is again similar to what we have learned in the class. Rather than keeping the same learning rate for the gradient information, we decrease it towards the end, which let the particles search the high dimensional parameter space faster at start and converge with slow movements to the optimal point they find.
- **Decreasing Inertia:** Similar to the above experiment of decreasing learning rate, we believe it is meaningful to give more importance to random velocity each particle have at the start so they can explore this high dimensional space more. And towards the end we would give less importance to it to get all the particles again converged into a single global best point.
- **All in One:** In the end, we wanted to combine all these experiments in one single training to analyze how well our method could have been. It is also meaningful because when we use particle momentum individually we had the effect of the initial random velocity all the time, but when we use it together with decreasing inertia this effect would also diminish towards the end.

Below in figure 1, we have ran our corresponding algorithms in LSTM with our data but the results were not as expected:

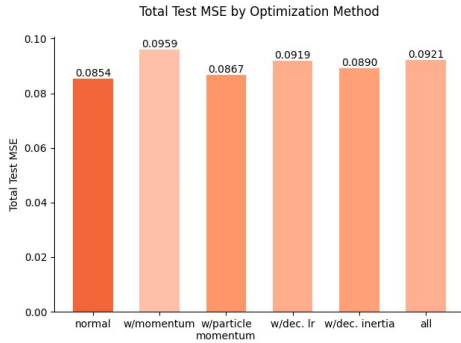


Fig. 1. Experiments done on Hybrid PSO approach

In the end, we had a better score on our normal version with simple addition of gradient information to the velocity, but the experimental methods are not that far off. Meaning not that our methods didn't work, but the randomization was probably in favor of normal version. To analyze this in detail we have ran multiple times and the results were indeed changing between 0.08 and 0.10 which is ideal for our problem.

B. Adding Gradient Once in X Turns:

This idea of ours was original, but we searched for possible implementations of Hybrid PSO as well. In one other version we have seen implementing or adding the gradient information only once in two rounds. To again understand the trade off between time and performance we have done the following experiment in figure 2 as a more general approach:

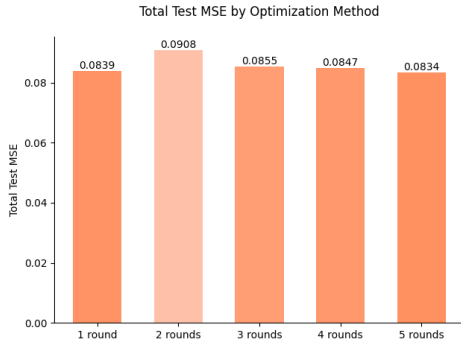


Fig. 2. The MSE when gradient is used once in X turns

The results were again surprising for us initially, we expected more of a distinction between using gradient information each turn versus using it once in 5 turns. But it seems like the effect of the global best position overpowers the gradient information of all the particles, which makes it more meaningful since it is the purpose of this gradient free algorithm. To observe the trade off even better, here is the time taken by each option from using it once in 1 round to once in 5 rounds in figure 3:

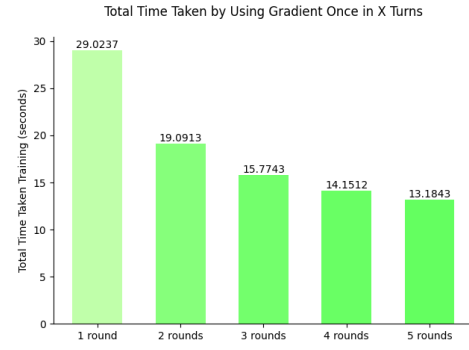


Fig. 3. Time taken by using gradient once in X turns

Seeing how the time decreases by increasing the round value, we tried to find the optimum point and as the most time taking part is calculating gradient for each particle it was rather fast to calculate for rounds from 10 to 100. Finally, we can observe this trade off as the following figure 4:

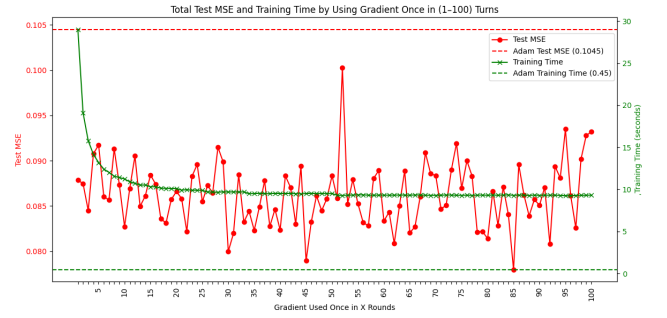


Fig. 4. Using gradient information for each particle once in X Turns

Looking at the figure 4, we can't say there is a strong correlation between the total mse and number of turns gradient information has been used, but still it should be noted the total MSE is in quite a short range which might be the reason of it. Also observing the green line, which is the time taken for each version, we might even be able to say that not using the gradient information can be even more effective for PSO.

After all these approaches, our main observation is that hybrid PSO was definitely better performing than Adam, but was worse in speed as can be seen in the figure 4 again (the straight red line on top of the figure is the MSE of Adam and the straight green line is the time taken for it as shown in the legend).

VII. CONCLUSION & FUTURE WORK

A. Conclusion

In this project, we tried to optimize the LSTM model to predict traffic congestion at districts of Istanbul level using different optimization techniques. Like Bayesian Optimization, Particle Swarm Optimization, its hybrid version, Adam along with many optimizations we made. Our results show that there are clear trade offs between the error and time taken with each method. We have tried to analyze these and here are the results:

As we can see in figure 5, Hybrid PSO achieves the lowest MSE error score followed by Adam and PSO. However, the most distinctively, the results show that Bayesian optimization has a significantly higher MSE error, indicating that this method has a lower prediction performance for our traffic congestion level.

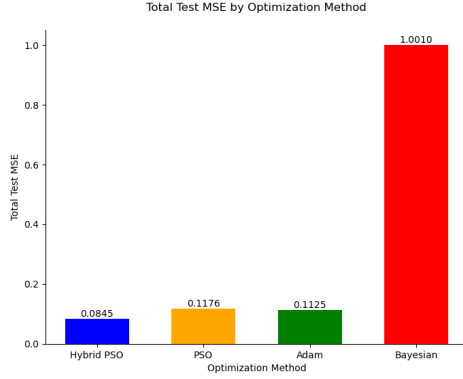


Fig. 5. MSE by Optimization Method

But when we look at figure 6 Bayesian required the longest optimization time followed by Hybrid PSO and normal PSO while Adam proved to be the fastest optimization technique among all the chosen techniques:

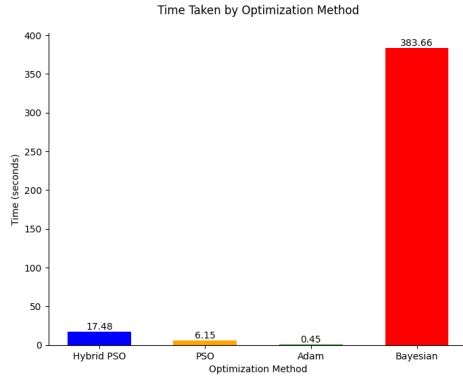


Fig. 6. Time Taken by Optimization Method

Going deeper into details, from the figure 4, we can see that the performance of hybrid PSO is comparably higher (the range of test scores doesn't start from 0) than Adam while requiring much higher times of training, a considerable trade off.

Another detail to mention would be the figure 1, which shows different experiments done upon the hybrid PSO approach to improve its accuracy. Our conclusion from this experiment was about how well the original PSO was originally structured. While we have used different methods like using gradient information, giving different importances to different parameters at different points of iterations, implementing momentum and more, we would still get similar results. This is because the personal and global best along with

the randomization factor from the initial velocity let's us find the optimal beautifully. Personal best directs us close to the best position we had been so far and that is what the gradient is trying to do as well. Global best directs us a single point to converge into and it is again similar to gradient information but in more of a general level. In the end, it seems like gradient information might not even be worth the resources spent on, especially in swarm optimization models.

Finally, these results show how important it is to select an optimization algorithm that provides the right balance between accuracy and efficiency and it might depend on users. We had so far concluded that while hybrid PSO provides the best prediction performance, Adam is a better choice because it has low MSE and the fastest convergence time!

B. Future Work

For the future work, there are lots of options. But we think several different methods could be tested to improve the performance and efficiency of the model as:

- **Hyperparameter Tuning:** We can perform hyperparameter tuning both for LSTM model and optimization algorithms we use, which can provide further improvements in accuracy and training speed.
- **Swarm Parallelization:** Knowing each swarm actually moves individually and act as a separate model alone, we believe that the points of swarm can be implemented in parallel and can share information within population about global best after each iteration.
- **Other Optimization Techniques:** While we have tested out several algorithms and experimented on them, testing and comparing additional optimization algorithms can still provide a better perspective on more suitable options for similar tasks.